

Effects of niobium alloying on microstructure, toughness and wear resistance of austempered ductile iron

Xiangru Chen^{a,*}, Long Zhao^a, Wei Zhang^c, Hardy Mohrbacher^{a,b}, Wenjun Wang^c, Aimin Guo^c, Qijie Zhai^a

^a School of Materials Science and Engineering, Shanghai University, Shanghai, 200444, China

^b NiobelCon bvba, Swaenebeecklaan, 2970, Schilde, Belgium

^c CITIC Metal Co., Ltd., Beijing, 100004, China

ARTICLE INFO

Keywords:

Ductile iron
Niobium alloying
Graphite nodularity
Austempering
Solubility
Carbide precipitation
Wear resistance
Toughness

ABSTRACT

Austempered ductile iron (ADI) is a heat-treated nodular iron variant that has for a given strength level much higher elongation than conventional iron with spheroidized graphite. This paper considers the effect of niobium addition on the graphite microstructure, bainite microstructure, bainite transformation process and properties such as hardness and impact toughness under given heat-treatment conditions. The resulting properties are significant with regard to the wear resistance of the material. The effects of niobium addition on the graphite morphology can be understood based on a detailed precipitation analysis of the niobium precipitation behavior in the liquid phase. Niobium influences the temperature and kinetics of pearlite as well as bainite transformation. Furthermore, niobium refines the bainite microstructure generated during austempering treatment. It was found that niobium addition in the range of 0.2–0.5 wt percent allows obtaining an optimum combination of hardness, impact toughness and wear resistance.

1. Introduction

Austempered ductile iron (ADI) represents a class of heat-treated nodular iron alloys offering at a given strength level almost twice the elongation of conventional nodular iron. Commercial ADI alloys are situated in the tensile strength range of 800–1600 MPa and are thus competing with ultra-high strength steel alloys. Due to its high graphite content, the density of ADI is around 10% lower than that of steel. Regarding the fatigue endurance, ADI is less notch sensitive than forging steel. The fatigue strength of ADI reaches the highest level when the heat treatment (austempering) results in a particularly high share of retained austenite [1]. On the contrary, the impact toughness of ADI is lower than that of forging steel but still about three times better than that of conventional nodular iron with [2]. The combination of high strength and elongation of ADI allows constructing comparably light components that can effectively compete with aluminum-based solutions. From an economical point of view ADI is more cost efficient than steel castings or forgings made from heat-treatable alloys. Therefore, ADI is an attractive material for lightweight components in machines, transportation equipment and vehicles [3]. Since many potential applications of ADI occur in capital goods, longevity and service reliability are key issues. In many relevant applications, longevity is not only

represented by fatigue endurance but also by wear resistance. Increasing the wear resistance of ADI is a central aspect of the present study. Besides optimizing the base microstructure of the ADI matrix, the addition of niobium, which forms hard niobium carbide (NbC) particles is considered for achieving this target. The dispersion of NbC particles in cast iron principally has the potential of increasing the wear resistance [4]. Niobium carbide is an extremely wear resistant compound [5,6]. Its density being 7.82 g/cm³ is very similar to that of liquid iron so that segregation of NbC particles by gravity is rather sluggish. After long holding without stirring NbC will finally segregate to the bottom of the ladle. The use of niobium for microstructural optimization of grey cast iron and wear reduction is already quite established [7–10]. Studies on niobium alloying in ductile iron (DI), however, are rather scarce. Higher niobium additions (0.25–0.65%) were found raising ductile-to-brittle transition temperature in Charpy tests, yet upper-shelf energy was not affected by niobium [11]. An early application used niobium alloying of around 0.5% in ductile iron for increasing the wear resistance of piston rings [12]. An elaborate work by Bedolla-Jacuinde et al. [13] indicated the formation of cube-shaped primary NbC particles which become rather large at high niobium additions (0.4–0.8% Nb). These contribute to modestly increasing the DI hardness by up to 6 HRC, yet on the expense of a significant loss of ductility. Neither nodule

* Corresponding author.

E-mail address: cxr16@shu.edu.cn (X. Chen).

<https://doi.org/10.1016/j.msea.2019.05.100>

Received 15 May 2019; Received in revised form 26 May 2019; Accepted 27 May 2019

Available online 28 May 2019

0921-5093/ © 2019 Elsevier B.V. All rights reserved.

count nor nodularity was affected by such niobium addition. A recent study by the present research group [14] has identified the effects of lower niobium additions of up to 0.11% on the microstructure of as-cast ductile iron. It was revealed that the 0.08% Nb resulted in a favorable combination of tensile strength and elongation. The morphology and amount of graphite nodules was nearly unaffected. Furthermore, niobium was found promoting pearlite formation and refining the spacing between ferrite and cementite lamellae. Another study considered the effect of larger Nb additions of up to 0.85% on the microstructure and properties of ductile iron [15]. The most favorable effect was found for a Nb addition of 0.23% resulting in a particularly high graphite nodule count. A large quantity of primary niobium carbides was formed in alloys containing more than 0.6% Nb. Cantera et al. [16] analyzed the effect of 0.2% Nb added to hyper-eutectic austempered ductile iron (ADI). The study lets conclude that niobium reduces the nodularity of graphite, yet it still remains on an acceptable level. Macroscopic NbC particles were observed in the microstructure. The mechanical properties of the niobium alloyed ADI, however, were generally inferior to those of the Nb-free reference alloy. A paper by Abdullah et al. [17] reports about experiments with alloying of 0.5% Nb in austempered ductile iron. The study comes to positive conclusions in terms of enhancing mechanical properties and corrosion resistance by the niobium addition. However, no explanation of the metallurgical effects causing the beneficial influence of niobium are given.

It is the aim of the present study to identify the effects of niobium alloying on the microstructure of ADI and to determine the optimum alloying range with regard to wear resistance. Variation of the niobium content between 0.2 and 1.1% is the only variable in the study. Microstructural analysis is performed in the as-cast ductile iron and after austempering using two different heat treatment schedules. The obtained ADI is subjected to mechanical and wear testing. The results are interpreted by a thorough contemplation of the metallurgical effects of niobium during solidification as well in solid state. The rich knowledge of niobium's physical metallurgical effects is thereby utilized.

2. Laboratory casting and heat treatment

The raw materials being pig iron, ferro-silicon, ferro-manganese, ferro-molybdenum, pure copper and ferro-niobium, were charged into an induction furnace to target chemical compositions according to Table 1. Molybdenum alloying is used to avoid pearlitic transformation during cooling to the austempering temperature. The addition is limited to 0.3% to avoid excessive molybdenum segregation to the intercellular regions. Further improved hardenability is achieved by a synergetic effect of alloying 0.5% copper in addition to molybdenum. The high silicon content inhibits carbide precipitation during austempering and thus helps stabilizing austenite. Manganese is also an effective hardenability agent but is usually kept low to develop maximum elongation and toughness in ADI. In the present experimental alloys, however, a high manganese addition is set, based on prior Thermocalc analysis. Accordingly, the increased manganese content helps avoiding the formation of undesired niobium-based eutectic phases and intermetallic phases (Laves phases). The chemical analysis of each sample was verified by spectral analysis using a cast sample with a dimension of

Table 1
Chemical composition of ADI experimental alloys.

Sample	Alloy addition in wt%								
	C	Si	Mn	Mo	Cu	Nb	P	S	CE
1	3.64	2.57	1.90	0.31	0.52	n.a.	0.04	0.017	4.51
2	3.69	2.34	2.13	0.31	0.54	0.21	0.04	0.018	4.48
3	3.56	2.60	2.09	0.30	0.55	0.55	0.04	0.014	4.44
4	3.59	2.55	2.00	0.31	0.54	0.68	0.05	0.019	4.46
5	3.64	2.63	1.94	0.31	0.54	1.06	0.05	0.018	4.53

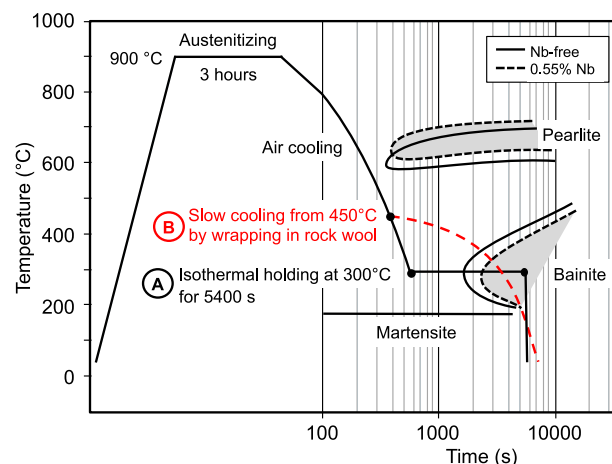


Fig. 1. Austempering treatment schedules practiced in the present study (positions of the pearlite and bainite phase fields indicated according to the CCT diagram).

Ø45 × 4 mm. Tapping of liquid iron was done in the temperature range of 1450–1500 °C. The casting temperature was adjusted to 1400 ± 20 °C and the liquid iron was cast into bars of Ø32 × 300 mm. For nodulizing the graphite FeSiMg10RE7 was added to an amount of 1.8%. The nodulizer was crushed into small pieces and placed on the bottom of the ladle where it was covered with inoculant (1.0% 75SiFe) and pearlite flux. Initially the ladle was filled to a level of 50–60% with liquid iron. The remaining iron was tapped when the melt in the ladle started boiling. Half of the inoculant covered the nodulizer before casting, the other half was added when iron liquid was cast. The flux was added after casting followed by de-slugging.

The as-cast ductile iron was heat treated by two different experimental schedules for obtaining an austempered microstructure (Fig. 1). In both treatment schedules austenitizing was performed at 900 °C for 3 h followed by air cooling. In schedule A, the material was held isothermally for 1.5 h at 300 °C in an electric resistance furnace and subsequently quenched to room temperature.

In schedule B, air cooling was interrupted at 450 °C and the material wrapped in rock wool allowing continuous cooling at slow rate. The pearlite and bainite phase fields, which were determined from prior dilatometer experiments are also indicated in this diagram. Both heat treatment schedules avoid the pearlite phase field and provide a reasonably long residence time in the bainite phase field.

3. Results

3.1. Characterization of the as-cast microstructure

Samples of all cast analyses were prepared for metallographic analysis by polishing and etching with 4% Nital solution. Optical micrographs of 100x magnification were quantitatively analyzed using Image-Pro Plus (Version 6.0) software to reveal the morphology and amount of graphite nodules. The procedure used for characterizing nodularity, graphite nodule count, and nodule roundness have been previously described in Ref. [14].

Fig. 2 represents the graphite morphology for all chemical analyses in the as-cast state. It is obvious that with increasing content of niobium the morphology of graphite deteriorates. Graphite nodularity is above 90% for niobium additions of up to 0.2% and slightly deteriorates at 0.55% Nb. Further increasing niobium additions result in a severe loss of nodularity to values below 80% (Fig. 2). Similarly, the number of graphite nodules per unit area is decreasing from a level of around 180 mm⁻² to below 120 mm⁻² with increasing niobium content (Fig. 2). The observations indicate a slight degradation of graphite characteristics when niobium is added in the range of 0.20–0.55%.

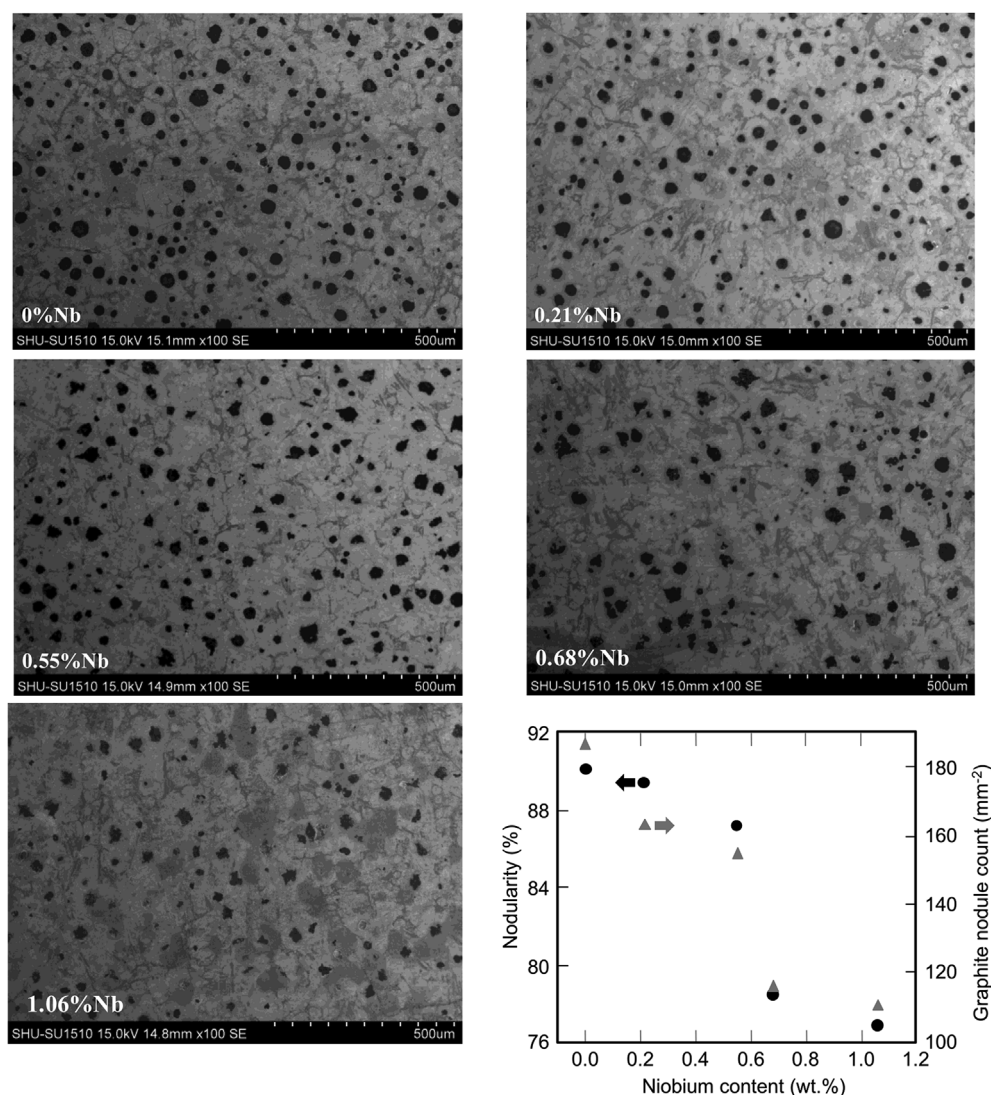


Fig. 2. Graphite morphology in the as-cast ductile iron for different niobium contents and quantitative evaluation of the nodularity and nodule count.

However, the nodularity is still fulfilling the minimum requirements of ASTM A536. Above a niobium content of 0.55% the degradation of graphite nodules is severe.

Niobium has a strong affinity to carbon resulting in the formation of NbC particles. Niobium carbide particles in the as-cast samples were analyzed using scanning electron microscopy (SEM). Representative SEM micrographs are shown in Fig. 3. The alloy containing 0.21% Nb comprises only few carbide particles having a size between 1 and 2 μm. With increasing niobium content, the NbC particle size increases to around 4 μm. Besides, the amount of NbC particles increases as well. Most of the NbC particles have a cubic morphology. For the largest niobium additions (0.68% and 1.06%) the NbC particles are aggregating to clusters.

The chemical composition of the particles was verified using micro-beam EDX. The spectra exhibit large Nb peaks as well as smaller peaks indicating the presence of Ti, Si, Mg and O. It is reasonable to assume that NbC particles nucleated on oxide precursors of Ti, Si or Mg present in the liquid phase. Niobium carbide particles precipitating after solidification have very small size, typically less than 100 nm. Such precipitates are not visible in the SEM micrographs of Fig. 3. Their detection would require TEM analysis.

3.2. Characterization of the austempered microstructure

The alloy concept has a significant influence on the phase transformation characteristics during an austempering treatment. The influence of niobium on the phase transformation behavior of the present alloys was analyzed by dilatometer tests. In this procedure, as-cast samples were heated to 900 °C with a heating rate of 10 °C/s and held at austenitizing temperature for 5 min. Subsequently, the phase transformation characteristics were recorded in a dilatometer using cooling rates ranging from 0.03 °C/s to 20 °C/s. Pearlite formation is suppressed for cooling rates above 1 °C/s and martensite starts forming in the temperature range of 150–200 °C. The phase field characteristics of pearlite as well as bainitic ferrite formation are listed in Table 2. Respective CCT curves have been recorded for the reference alloy and for the alloy containing 0.55% Nb. The obtained phase fields are also indicated in Fig. 1 with respect to the heat treatment schedule.

It is obvious that the addition of niobium to the alloy raises the nose temperature of the pearlite phase field. Oppositely, niobium decreases the nose temperature of the bainitic ferrite phase field, thus expanding the temperature gap between the pearlite and bainitic ferrite transformation. More importantly, niobium delays the incubation period for either phase.

Metallographic specimens of the austempered material were prepared by Nital etching (alcohol + 4% nitric acid). The developed

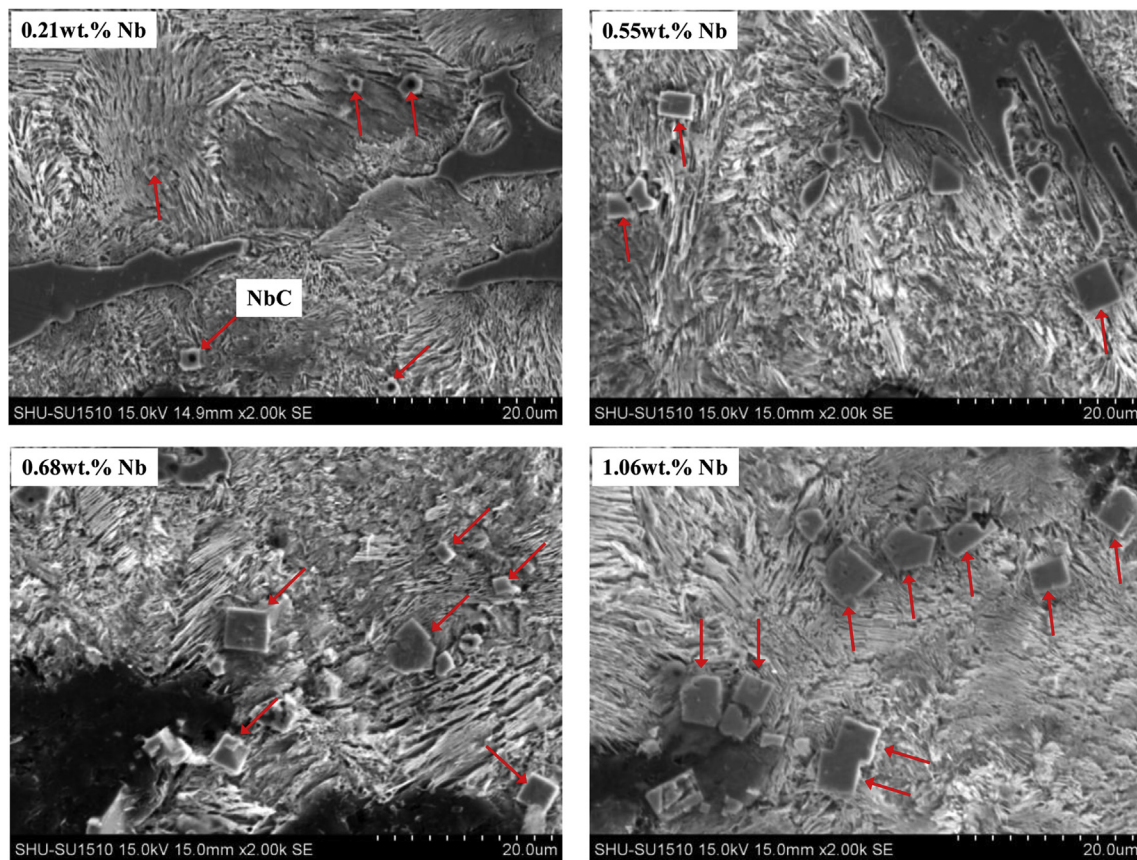


Fig. 3. Scanning electron micrographs of ADI samples with varying niobium content indicating the size, amount and morphology of niobium carbide particles (arrows).

Table 2

Characteristics of phase formation after austenitizing Nb-free and Nb-alloyed ductile iron at 900 °C.

Nb content		0 wt%	0.55 wt%
Nose temperature	Pearlite	598 °C	621 °C
	Bainitic ferrite	303 °C	294 °C
Critical cooling rate	Pearlite	0.9 °C/s	0.7 °C/s
	Bainitic ferrite	0.4 °C/s	0.3 °C/s
Incubation period	Pearlite	332 s	392 s
	Bainitic ferrite	1623 s	2022 s

microstructures were characterized using scanning electron microscopy and are shown in Fig. 4. Both microstructures consist mainly of bainite with acicular morphology. The bainite microstructure of Nb-free ADI (Fig. 4a) appears to be coarser than the one of the niobium alloyed variant (Fig. 4b). Also the degraded nodularity is clearly reflected in this micrograph of the niobium-alloyed ADI.

Quantitative analysis of the amount of bainite phase in the microstructure was performed after color etching of the micrographs. The etchant consisted of 2 g potassium metabisulfite and 2 g of ammonium bi-fluoride in 100 ml distilled water (Beraha-Martensite agent). The samples were dried immediately after complete colorization of the sample surface (Fig. 5). Beraha-Martensite etching develops martensite with a blueish contrast and bainite with a reddish-brown contrast,

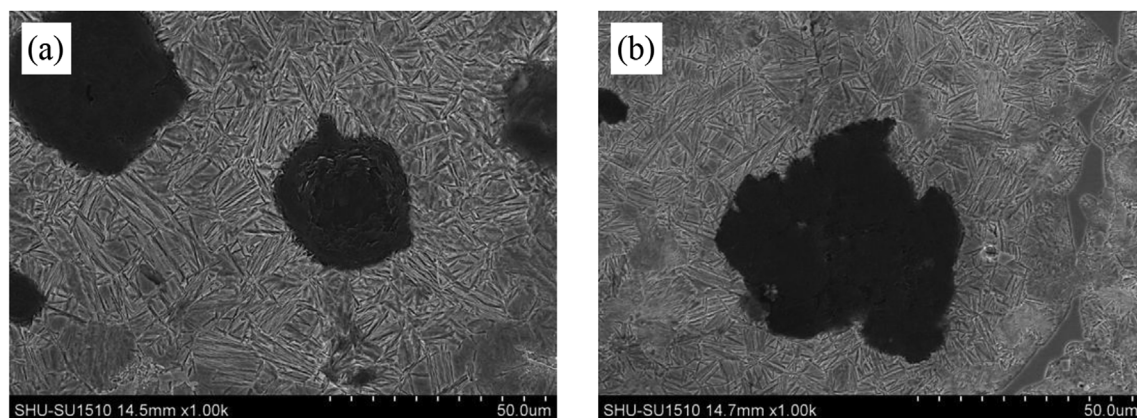


Fig. 4. Scanning electron micrograph (Nital etching) after heat treatment schedule A of reference ADI (a) and 0.55% Nb alloyed ADI (b).

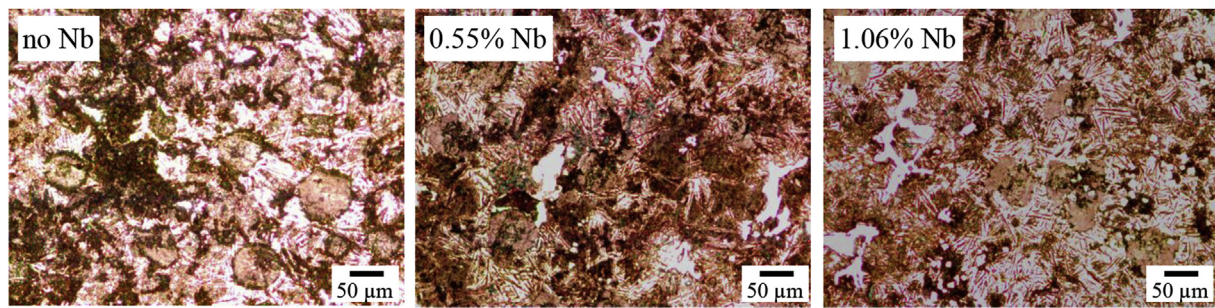


Fig. 5. Color etching with Beraha-Martensite agent of different ADI variants produced by heat treatment A. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

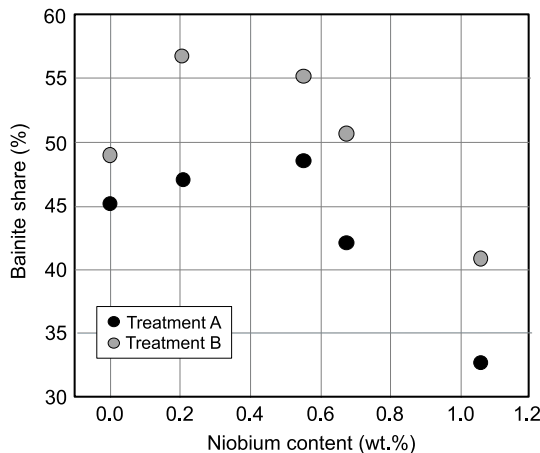


Fig. 6. Effect of niobium alloying and heat treatment schedule on the bainite share of austempered samples.

while the residual austenite and carbides appears as light contrast [18]. The bainite content was evaluated by image analysis software. The result of this quantitative analysis is shown in Fig. 6. The bainite share is slightly larger for all alloys heat treated by schedule B. It reaches a value of 57% in the best case (schedule B, 0.21% Nb). Alloying of niobium up to 0.55% increases the bainite share for both heat treatment schedules. When the niobium content exceeds this level, the bainite share drops. In the worst case (schedule A, 1.06% Nb) it is reduced to 33%.

3.3. Mechanical properties after austempering

Hardness, toughness and wear behavior have been characterized to identify the influence niobium addition after austempering the samples. The hardness was measured by a Rockwell tester using 100 kg indentation load. The hardness values shown in Fig. 7 represent the average value of five individual measurements. Comparing the Nb-free material with the 0.2% Nb added alloy indicates a hardness increase by 3 HRC. Further increasing niobium content results in a more moderate hardness gain. For heat treatment schedule A, the hardness even decreases again when the niobium content exceeds 0.6%. However, it is still higher as compared to the Nb-free material.

Toughness was tested on a JB-6 type impact testing device using an un-notched specimen of dimension $10 \times 10 \times 55 \text{ mm}^3$. The average values of three individual tests are shown in Fig. 8. Toughness is consistently higher for material heat treated by schedule A. After heat treatment schedule A toughness slightly improves with increasing niobium content up to a level of 0.55%Nb. For higher niobium additions toughness decreases rapidly. After heat treatment schedule B toughness starts deteriorating already at a niobium content lower than 0.55%.

Tribological testing was carried out on the MMS-2A friction and

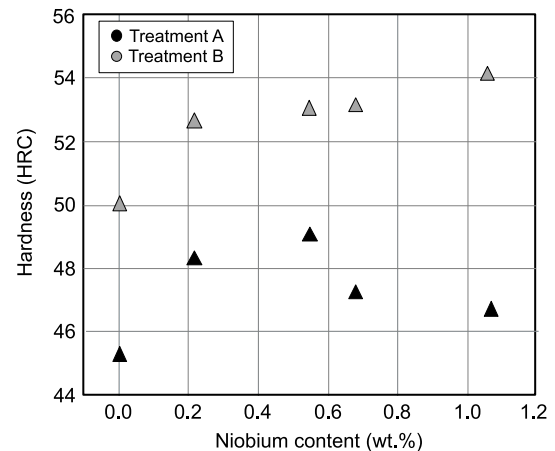


Fig. 7. Effect of niobium addition and heat treatment on Rockwell hardness (HRC).

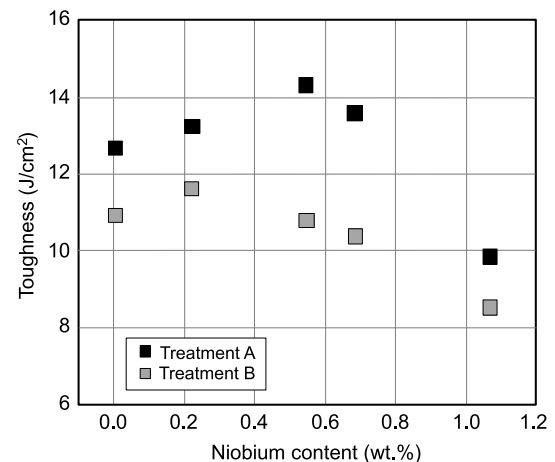


Fig. 8. Effect of niobium addition and heat treatment on Charpy impact toughness.

wear testing machine using a block-on-ring dry-frictional setup. The block of size $28 \times 7 \times 4 \text{ mm}^3$ represented the ADI material. The material used for the friction ring of size $\varnothing 40 \times 10 \text{ mm}$ was alloy steel heat-treated to a Rockwell hardness of 50 HRC. The surfaces of these tribo-couples were ground by 1000 grid paper and cleaned with alcohol before and after wear testing executed for a duration of 30 min at an applied contact load of 80 N and a rotation speed of 200 rpm. Prior to the actual wear testing, a running-in period of 10 min duration was practiced. The friction coefficient was recorded online at intervals of 10 s. The weight loss of the sample before and after the wear test was determined with an electronic balance (ESJ205-4) having a

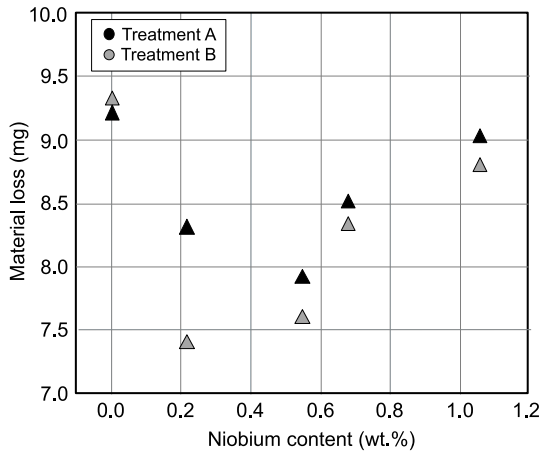


Fig. 9. Effect of niobium addition and heat treatment on material loss of ADI in block-on-ring sliding tests against 50 HRC treated alloy steel.

measurement precision of 0.1 mg.

The influence of the niobium addition on the weight loss is shown in Fig. 9. A significant reduction in weight loss is observed with increasing niobium content towards a value of 0.55%. The samples with niobium additions of 0.68% shows an abrupt increase in weight loss. The growth in weight loss continues for material containing 1.06% Nb showing a performance that is marginally better than the Nb-free material. The best performance is observed for the material containing 0.21% Nb treated by schedule B. This sample has neither the highest hardness nor the best toughness, but it comprises the highest bainite share. It has to be noted materials obtained by heat treatment A are all softer than the counter-body (50 HRC) while those produced by heat treatment B are all harder. However, most of the samples at a given niobium addition show very similar weight loss for heat treatment schedules A and B. Therefore, hardness appears not to play a dominating role regarding the wear resistance under the current circumstances. The assumption that simply enhancing the amount of NbC particles in the material should increase the wear resistance is contradicted by the current results. The on-line recorded steady-state coefficient of friction in all tribo-couples remained in a narrow range of 0.64–0.66. The alloys containing 0.21 and 0.55% Nb showed the lowest friction values within this range.

4. Discussion

Niobium alloyed to iron-based materials has pronounced physical metallurgical effects in both constitutions, as solute and as precipitate. Typical prominent effects are grain size control (refinement), transformation delay and interaction with carbon. The process of producing ADI incorporates potential interaction of niobium during solidification and the multiple subsequent phase transformations in solid state. Therefore, it is of key importance to analyze the status of niobium, solute or precipitate, during all these stages. Interpretation of the observed results will be done on this basis.

4.1. Niobium solubility in liquid and austenite phase

Thermodynamic calculations analyzing the Gibbs free energy of formation allow predicting the primary precipitation of niobium in liquid phase for the current ADI alloy composition. In this approach [19,20], a multicomponent ideal solution was considered using available activity coefficients and activity interaction coefficients of the alloying elements listed in Table 1. The resulting formula defines the temperature (in Kelvin) at which a given niobium addition precipitates in the liquid phase as:

$$T = (143623 - 10040 [\text{Nb}]) / (86.39 - 8.314 \ln [\text{Nb}]) \quad (1)$$

Table 3

Calculated limit temperature for liquid phase solubility of niobium in current iron alloys.

Nb alloy content (wt.%)	0.21	0.55	0.68	1.06
NbC formation temperature (°C)	1154	1239	1254	1275

Equilibrium formation temperatures of NbC calculated on this basis are displayed in Table 3 for the four investigated niobium additions used in this study. As expected, the NbC formation temperature rises with increasing niobium content. This analysis strictly holds for equilibrium conditions. When cooling down from the fully solute state, nucleation and growth of primary NbC particles requires time. In practical applications, depending on the cooling rate and the presence of heterogeneous nucleation sites (e.g. oxide particles), deviations from the calculated value are possible. If primary NbC precipitation is incomplete, solute niobium accumulates in interdendritic liquid pools or in intercellular regions where it finally forms eutectic carbides [21]. The current alloys having a CE of around 4.5 are all hyper-eutectic. Complete primary precipitation of the highest niobium content considered (1.06% Nb) withdraws 0.13% carbon from the melt. Therefore, all alloys of this study will still be hyper-eutectic even assuming complete primary niobium precipitation.

Experimental data indicated that for the present Nb-free alloy the eutectic temperature is 1132 °C while for the alloys containing niobium it is reduced to 1125 °C. At this temperature 0.17% Nb can remain solute in the liquid phase according to equation (1).

The solubility of niobium in austenite can be calculated using an equation proposed by Ohtani et al. [22], which is applicable for austenite with high carbon content:

$$\log [\text{Nb}][\text{C}] = -\frac{4880}{T} + 1.18 + [\text{C}] \left(\frac{1590}{T} - 0.1 \right) \quad (2)$$

In this equation T represents the temperature in Kelvin while [C] is the actual carbon concentration in austenite and [Nb] the resulting soluble niobium in austenite, both in weight percent. The actual carbon content at eutectic temperature for the current iron alloys is estimated to be 1.2%. Accordingly, equation (2) predicts a maximum niobium solubility in austenite of 0.07%.

Both solubility limit lines for niobium are graphically displayed in Fig. 10. Comparing niobium solubility in liquid and austenite phase at eutectic temperature indicates a discrepancy of 0.1% Nb that is neither soluble in liquid nor in austenite. Accordingly, this amount of niobium must precipitate during eutectic solidification when the original addition is at least 0.17% Nb. This is the case for all alloys tested in this

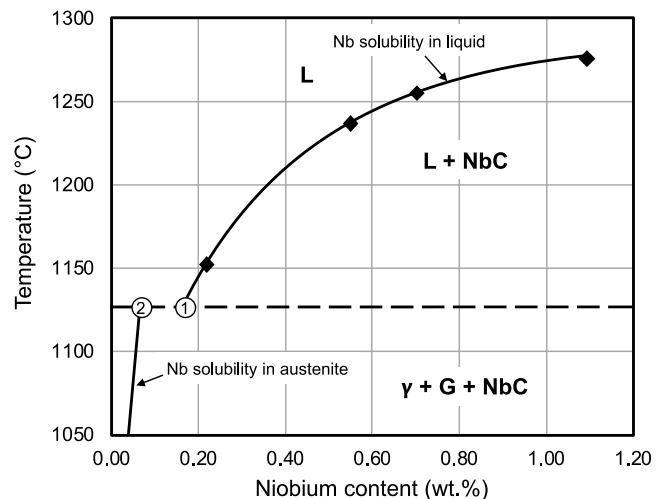


Fig. 10. Different stages of niobium precipitation in hyper-eutectic ADI cast iron; niobium solubility limits in liquid and austenite.

study. The precipitation of niobium during the solidification reaction can have an inoculating effect for graphite. The lattice disregistry between NbC and graphite is, depending on the crystalline orientation relationship, 7.2%, 8.2% and 11.2%, respectively. Thus, NbC has the potential of serving as heterogeneous nucleation site for graphite. Large primary NbC precipitates or clusters can, however, promote preferred directional growth of graphite and thus result in low nodularity. This was observed for the higher niobium alloyed samples in this study. The depletion of carbon from the melt by an increasing amount of primary NbC is reflected in the reduced graphite nodule count.

4.2. Niobium effects on the solid-state phase formation

After complete solidification, niobium solubility in austenite is rapidly declining with decreasing temperature. However, precipitation kinetics of NbC in austenite is very sluggish. The orientation relationship between NbC and austenite is cube-on-cube having a lattice disregistry of 25%. Thus, nucleation of NbC is obstructed and a substantial amount of niobium remains in solid solution outside the equilibrium. Solute niobium delays the transformation from austenite to ferrite or pearlite by solute drag on the phase boundary and reduced carbon diffusivity [10]. These effects result in significant refinement of the pearlite lamellar spacing [14]. The solubility of niobium in the bcc-phase is extremely low and the lattice disregistry between [010]-Fe and [011]NbC is only 10%. This results in precipitation of ultra-fine particles, typically having the size of a few nanometers.

Upon re-austenitizing at 900 °C, these NbC nano-precipitates are mostly stable and exert an anti-coarsening effect on the austenite grains by boundary pinning. This effect is standardly used in the production of quench & temper steels [23] or case carburizing treatments [24]. At 900 °C only about 0.01% Nb can re-dissolve according to equation (2). The austenite grain size control and the small amount of solute niobium have an influence for the subsequent transformation behavior upon down-cooling to the austempering temperature (Table 2). The larger total austenite grain boundary surface area and increased number of triple points between austenite grains facilitate pearlite nucleation in the niobium alloyed material. However, since solute niobium reduces the diffusivity of carbon in austenite and exerts solute drag on the phase boundary, the onset of pearlite formation is retarded. Considering bainite formation, solute niobium enhances the stability of austenite and reduces the free energy of phase transformation. This results in a reduced transformation temperature and a prolonged incubation period.

The refined austenite microstructure provides more nucleation sites for bainite, thus refining the bainite packet structure. It has been reported that typically four bainite packets develop within each prior austenite grain [25]. Consequently, the shorter bainite needles should develop in fine-grained austenite. Niobium alloying can provoke this effect.

Bainite formation during the austempering treatment occurs by displacive growth of ferrite needles, which are initially supersaturated with carbon [26]. Subsequently, carbon diffuses into neighboring austenite. However, the isothermal transformation will stop when the carbon content in the austenite phase being in contact with the ferrite needle has saturated. Ferrite formation can only continue if cementite precipitation occurs. High alloying content of silicon, however, obstructs the precipitation of cementite. A preferred nucleation site for ferrite needles is the graphite-to-austenite interface [27] since graphite nodules act as carbon sink [28]. Accordingly, bainite formation is concentrated around graphite nodules and less pronounced in the intercellular regions.

The reduced nodule count at higher niobium levels (Fig. 3) can thus be considered as one reason for the reduced bainite share observed in these alloys. Furthermore, large primary NbC particles dispersed in the austenite matrix can act as barriers obstructing the growth of bainite. However, for lower niobium addition the bainite share slightly

increases. This can be due to a higher density of boundary nucleation sites in the refined austenite and to the presence of nano-sized NbC precipitates also acting as nucleation sites [29]. Furthermore, diffusion distances for carbon are shorter in the grain refined microstructure facilitating carbon partitioning. The effects obstructing bainite growth obviously start dominating in the present alloys above a niobium content of 0.55% (Fig. 6).

4.3. Implications on the mechanical properties

The mechanical properties of austempered ductile iron are influenced by the individual phase components, their morphology and their spatial distribution. Therefore, the interpretation of the observed mechanical behavior is rather complex. With regard to niobium, it is obvious that the best mechanical properties are achieved in the alloying range of 0.2%–0.6% Nb. This is in line with most of the results reported in literature on niobium alloyed ductile iron. Higher niobium additions do not result in further improvement but can even lead to a deterioration of mechanical properties. The degradation of graphite nodularity as well as the presence of large and clustered primary NbC particles are considered being majorly responsible for the reduced toughness at high niobium additions. Also, the wear resistance is not benefitting from the presence of an increased amount of coarse carbide particles.

A high bainite share is found to be generally beneficial for the mechanical properties. Toughness and hardness improve with the amount of bainite until detrimental microstructural features become dominant at higher niobium content. The two different heat treatments practiced in this study have also an impact on the bainite share. Treatment schedule B produces more bainite than schedule A. However, the mechanical characteristics allow to conclude that the remaining phase after both treatments is different. This phase appears to be softer (hardness data) and more ductile (toughness data) after schedule A.

The main motivation of this study was to investigate whether niobium alloying can improve the wear resistance of ADI. Fig. 9 suggests that an optimum wear resistance is achieved in the alloying range of 0.2%–0.5% Nb. In an attempt to unravel the microstructural contributions to wear resistance, the material loss data of Fig. 9 are normalized by the respective value of the Nb-free material for both heat treatment schedules. These data are then plotted in function of the bainite share (Fig. 11). The plot indicates a good linear correlation of these data. Hence, it can be concluded that increasing bainite share improves the wear resistance. The improvement has a magnitude of 20% over the variations in bainite share (33%–56%) realized in this

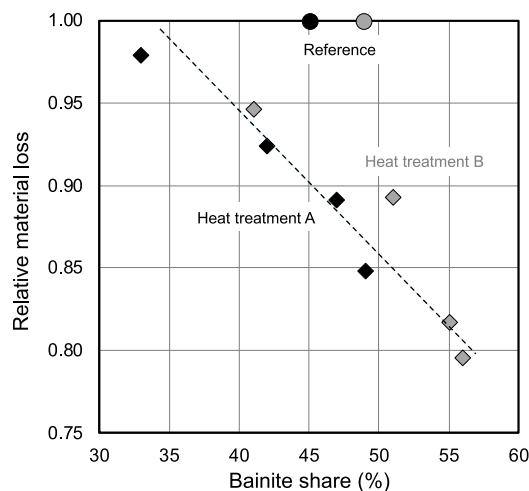


Fig. 11. Relative material loss of niobium alloyed material as a function of bainite share (the dashed line represents a linear fit of all samples with Nb addition (diamond symbol)).

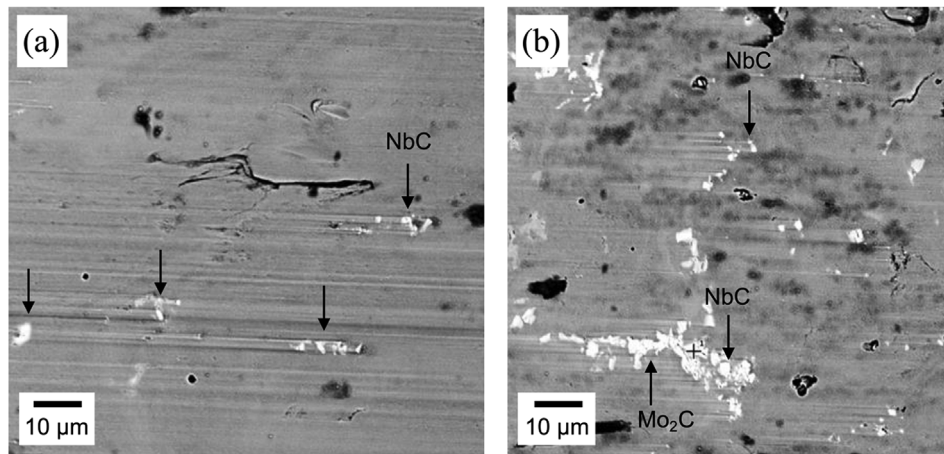


Fig. 12. Scanning electron micrographs of wear tracks in 0.55% Nb alloyed ADI indicating (a) the obstruction of ploughing by dispersed NbC particles; (b) large-sized eutectic carbides formed by molybdenum segregation in the intercellular region.

study. Alloying of niobium influences the bainite share and thus contributes indirectly to this improvement. The hardness that is generally higher after heat treatment schedule B (Fig. 7) does not make a significant contribution to the wear resistance. However, one can also observe a direct impact of niobium on the wear resistance. Comparing the normalized material loss of the Nb-free reference material with that of a Nb-alloyed material having the same bainite share, an improvement of wear resistance in the order of 10% is seen. This effect can be attributed to hard NbC particles embedded in the material. The contribution of these NbC particles is understood as stopping abrasive third body particles from ploughing the ADI surface. Scanning electron microscopy of wear tracks demonstrates this effect in Fig. 12. Energy dispersive analysis (EDX) indicated that the embedded particles at which the surface scratches are interrupted indeed consist of niobium carbide (Fig. 12a). In the intercellular region, larger eutectic carbides are observed consisting mainly of molybdenum carbide due to the pronounced segregation of molybdenum to this region (Fig. 12b). The NbC particles are more homogeneously spread and also appear within the eutectic cells indicating that they must have been formed before eutectic solidification. One could suppose that also the graphite characteristics might have an influence on the wear behavior. However, the friction data showing very little difference between the alloys do not support this assumption.

5. Conclusions

The effect of niobium alloying to austempered ductile iron was investigated with the aim of improving wear resistance. It was found that niobium influences the microstructure during all individual stages of the ADI processing and is thus also modifying the mechanical properties of the final product.

1. For the present base alloy composition, primary NbC precipitation in the liquid phase is expected for niobium additions above 0.17%. No precipitation in the liquid phase is expected for niobium additions below 0.07%. Niobium in the range of 0.07–0.17% must precipitate during eutectic solidification. These limit values depend primarily on the carbon and silicon contents in the alloy. The limit values can be calculated for other alloy compositions using available solubility equations.
2. NbC present in the liquid phase acts as heterogeneous nucleation site for graphite. NbC particles can assist graphite inoculation. Large NbC particles or particle clusters deteriorate graphite nodularity. The carbon consumption caused by NbC formation reduces the graphite nodule count.
3. Niobium remains largely in solid solution during down-cooling of

austenite. Solute niobium delays the phase transformation and refines the pearlite lamellar structure. Niobium acts as a pearlite stabilizer.

4. Niobium precipitates in the bcc-phase as nano-sized particles enhancing strength and hardness. These particles obstruct austenite grain coarsening during the re-austenitizing treatment. A small amount of niobium is re-dissolved during austenitizing. This solute niobium retards pearlite and bainite formation. The critical cooling rate for avoiding pearlite formation is thus reduced.
5. The refined austenite grain structure and the NbC nano-particles are nucleation sites for bainite formation. Niobium thus increases the bainite share. The isothermal holding time has to be optimized by separate experiments for obtaining the highest possible bainite share.
6. A higher bainite share is beneficial to the mechanical properties of ADI. However, high niobium addition causing detrimental microstructural features can counteract the positive effect of bainite.
7. The combination of increased bainite share and homogeneously dispersed primary NbC particles substantially enhances the wear resistance of ADI.

The current study has demonstrated that the optimum addition range for niobium in this hyper-eutectic ADI alloy is between 0.2% and 0.5% Nb.

Acknowledgements

The authors acknowledge the financial support from the CITIC-CBMM Nb Steel Research and Development Fund (No. 2013-D069) and from the Natural Science Foundation of China (contract no. 51404150).

References

- [1] M. Grech, J.M. Young, Impact properties of Cu-Ni austempered ductile iron, *Cast Metals* 1 (2) (1988) 98–103.
- [2] R. Martinez, R. Boeri, J. Sikora, Impact and Fracture Properties of ADI - Comparison with 4140 Steel, American Foundrymen Society Congress, 1998 Paper No. 98-008.
- [3] R.A. Harding, The production, properties and automotive applications of austempered ductile iron, *Kov. Mater.* 45 (2007) 1–16.
- [4] H. Mohrbacher, D. Jarreta, Technology, properties and applications of niobium carbide reinforced iron and steel alloys, in: H. Mohrbacher (Ed.), *Fundamentals and Applications of Mo and Nb Alloying in High Performance Steels*, vol. 2, 2015, pp. 227–256.
- [5] M. Woydt, H. Mohrbacher, Friction and wear of binder-less niobium carbide, *Wear* 306 (2013) 126–130.
- [6] M. Woydt, H. Mohrbacher, The use of niobium carbide (NbC) as cutting tools and for wear resistant tribosystems, *Int. J. Refract. Metals Hard Mater.* 49 (2015) 212–218.
- [7] E. Campomanes, R. Goller, Effects of Nb addition on the properties and structure of

- gray iron, AFS Trans. 73-8 (1973) 122–125.
- [8] S.V. Subramanian, A.J. Genualdi, AFS Trans. 96–138 (1996) 995–1001.
 - [9] W. Zhou, H. Zhu, D. Zheng, H. Zheng, Q. Hua, Q. Zhai, China Foundry 8 (1) (2011) 36–40.
 - [10] H. Mohrbacher, Q. Zhai, Proceedings of Materials Science & Technology Conference, ASM Int., Columbus, OHIO, 2011, pp. 434–445.
 - [11] M.J. Fallon, BCIRA Journal, (July 1984), pp. 274–279.
 - [12] J. Vatauvuk, J.R. Mariano, JOM 44 (Issue 1) (1992) 13–14.
 - [13] A. Bedolla-Jacuinde, E. Solis, B. Hernandez, Effect of niobium in medium alloyed ductile cast iron, Int. J. Cast Metals Res. 16 (5) (2003) 481–486.
 - [14] X. Chen, J. Xu, H. Hu, H. Mohrbacher, M. Kang, W. Zhang, A. Guo, Q. Zhai, Effects of niobium addition on microstructure and tensile behavior of as-cast ductile iron, Mater. Sci. Eng. A 688 (2017) 416–428.
 - [15] T. Nogueira Fonseca Souza, R. Alves Perilo Sousa Nogueira, F. Saldanha Franco, M. Paulino Aguilard, P. Cetlina, Mechanical and microstructural characterization of nodular cast iron (NCI) with niobium additions, Mater. Res. 17 (5) (2014) 1167–1172.
 - [16] D. H. Munhoz Cantera, O. Preti, D. J. Horst, R. De Almeida Vieira, C. A. Duvoisin, Am. J. Eng. Res. , Volume-7, Issue 7, pp. 181–193.
 - [17] B. Abdullah, S. K. Alias, A. Jaffar, A. Abd Rashid, A. Ramli, Mechanical properties and microstructure analysis of 0.5% niobium alloyed ductile iron under austempered process in salt bath treatment, Proc. International Conference on Mechanical and Electrical Technology (ICMET 2010), pp. 610–614.
 - [18] J.M. Radzikowska, Metallography and Microstructures of Cast Iron, ASM Handbook vol. 9, Metallography and Microstructures, ASM International, Materials Park, Ohio, USA, 2004, pp. 565–587.
 - [19] Y. Liang, Codata Key Values for Thermodynamics, Northeastern University Press, Shenyang, 1993, pp. 508–510.
 - [20] Q. Zhai, Thermodynamical problems of trace elements in cast irons, Mod. Cast. Iron (01) (2001) 19–23.
 - [21] C.R. Loper, H.K. Baik, Influence of molybdenum and titanium on the microstructures of Fe-C-Cr-Nb white cast iron, AFS Trans. 89–97 (1984) 1001–1008.
 - [22] H. Ohtani, T. Tanaka, M. Hasebe and T. Nishizawa, Solubility of NaCl-type carbides (NbC, VC and TiC) in austenite, Proc. Of the Japan-Canada Seminar, Dec. 1985, P. J-7-1 – J-7-12.
 - [23] H. Mohrbacher, Property optimization in as-quenched martensitic steel by molybdenum and niobium alloying, Metals 8 (4) (2018) 234.
 - [24] Th Tobie, F. Hippenstiel, H. Mohrbacher, Optimizing gear performance by alloy modification of carburizing steels, Metals 7 (10) (2017) 415.
 - [25] C. Wang, X. D. S. Li, Y. Sun, P. Yang, Morphology and crystallography of ausferrite in austempered ductile iron, Metals 7 (2017) 238, <https://doi.org/10.3390/met7070238>.
 - [26] R.B. Gundlach, J.F. Janowak, S. Bechet, K. Röhrig, Transformation behavior in austempering nodular iron, in: H. Fredriksson, M. Hillert (Eds.), Proc. Of the 3rd Int. Symp. On the Physical Metallurgy of Cast Iron, Stockholm, vol. 34, MRS, 1984, pp. 399–409.
 - [27] G. Gumieny, Bainitic-martensitic nodular cast iron with carbides, Arch. Foundry Eng. 10 (Issue 2) (2010) 63–68.
 - [28] H. Nieswaag, J.W. Nijhof, Influence of silicon on bainite transformation in ductile iron, in: H. Fredriksson, M. Hillert (Eds.), Proc. Of the 3rd Int. Symp. On the Physical Metallurgy of Cast Iron, Stockholm, vol. 34, MRS, 1984, pp. 411–422.
 - [29] J. Kaleicheva, V. Mishev, Z. Karaguiozova, Effect of nanoadditives on the characteristics of bainitic transformation in spheroidal graphite cast irons, FDIBA Conf. Proc. Sofia 1 (2017) 97–100.